Calculus Assignment 4

1. Proof: Assume $f(m) > 0$ for some m and $f(n) < 0$ for some n, $m, n \in [a, b]$. Without loss of generality, assume that $m < n$. Since $f(x)$ is continuous on $[m, n] \subset [a, b]$ and $f(m) \cdot f(n) < 0$, Intermediate Value Theorem states that there exists $c \in (m, n)$ such that $f(c) = 0$. But $f(x) \neq 0$ for all $x \in [a, b]$. Contradiction! Hence $f(x)$ is strictly positive or strictly negative on $[a, b]$. □

2. (a) To find the velocity of the rock, we need to calculate the derivative of the height function with respect to time t, which is $v(t) = \frac{dH(t)}{dt} = 10 - 2(1.86)t = 10 - 3.72t$. So the velocity of the rock after 1 second is 6.28 m/s.

(b) From (a), we know $v(t) = 10 - 3.72t$. Solve $H(t) = 10t - 1.86t^2 = 0$ and we get $t = 0$ or $t \approx 5.3763$ s. Hence the velocity of the rock when $t = a$ is $10 - 3.72a$ m/s if $a \in [0, 5.3763]$ or 0 m/s otherwise

(c) When the height of the rock is zero again, the rock hits the surface. We get $H(t) = 0$ when $t = 0$ and $t \approx 5.3763$s from (b). Hence the rock will hit the surface after about 5.3763 s.

(d) From (b) and (c), we know that the rock will hit the surface with the velocity $v(5.3763) = 10 - 3.72 \times 5.3763 \approx -10$ m/s. The negative sign here indicates the rock is moving downward.

3. (a) $V(a+h) - V(a)$ represents the volumn between the depth a and $(a+h)$.

(b) Since $A(a) \geq A(a+h)$, we have $A(a+h)h \leq V(a+h) - V(a) \leq A(a)h$.

(c) Following (b), we get $\frac{A(a+h)h}{h} \leq \frac{V(a+h)-V(a)}{h} \leq \frac{A(a)h}{h}$, namely: $A(a+h) \leq \frac{V(a+h)-V(a)}{h} \leq A(a)$, which indicates $\frac{V(a+h)-V(a)}{h}$ approaches $A(a)$ as h tends to 0.

4. (a) $f'(4) = \lim_{h\to 0} \frac{\sqrt{4+h} - \sqrt{4}}{h} = \lim_{h\to 0} \frac{4+h-4}{h(\sqrt{4+h}+\sqrt{4})} = \lim_{h\to 0} \frac{1}{\sqrt{4+h}+\sqrt{4}}$

$= 1/4$. Also $f(4) = 2$. So the line tagent to the graph of f at $x=4$ is $y-2 = 1/4(x-4)$, i.e. $y = 1/4\,x + 1$.

(b) $f'(2) = \lim_{h\to 0} \frac{m(2+h)^2 + k(2+h) - 4m - 2k}{h} = \lim_{h\to 0} \frac{4mh + mh^2 + kh}{h}$

$= \lim_{h\to 0} \frac{4m + mh + k}{1} = 4m + k$. Also $f(2) = 4m + 2k$. So the line tagent to the graph of f at $x=2$ is $y - (4m+2k) = (4m+k)(x-2)$, i.e. $y = (4m+k)x - 4m$.

(c) $f'(-1) = \lim_{h\to 0} \frac{(-1+h)^3 - (-1)^3}{h} = \lim_{h\to 0} \frac{(-1)^3 - 3h^2 + 3h + h^3 + 1}{h}$

$= \lim_{h\to 0} \frac{-3h + 3 + h^2}{1} = 3$. Also $f(-1) = -1$. So the line tagent

to the graph of f at $x=-1$ is $y+1=3(x+1)$, i.e. $y=3x+2$.

5. Proof: Assume $f(x)$ is a function that is not continuous at a but differentiable at a. However, as shown in the lecture slides, differentiability indicates continuity, so $f(x)$ is continuous at a. Contradiction! Hence a function that is not continuous at a can not be differentiable at a. □

6. (a) $y'_+(0) = \lim_{\Delta x\to 0^+} \frac{(\Delta x)e^{-\frac{1}{\Delta x}} - 0}{\Delta x} = \lim_{\Delta x\to 0^+} e^{-\frac{1}{\Delta x}} = 0$

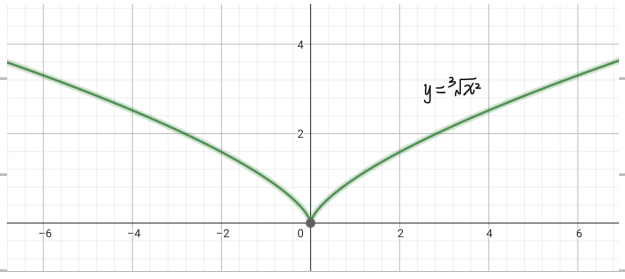
(b) $y'_+(0) = \lim_{\Delta x\to 0^+} \frac{(\Delta x)^2 \sin\frac{1}{\Delta x} - 0}{\Delta x} = \lim_{\Delta x\to 0^+} \Delta x \sin\frac{1}{\Delta x} = 0$

$y'_-(0) = \lim_{\Delta x\to 0^-} \frac{\cos\Delta x - \Delta x - 1 - 0}{\Delta x} = \lim_{\Delta x\to 0^-} \frac{\cos\Delta x - 1}{\Delta x} - 1 = -1$

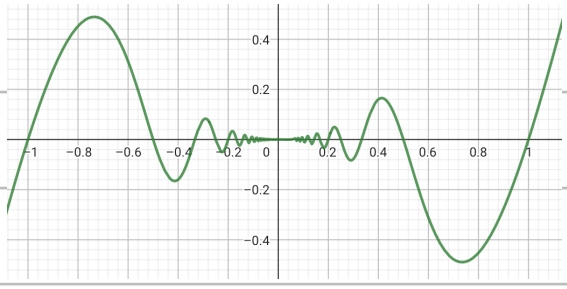
Since $y'_+(0) \neq y'_-(0)$, this function is not differentiable at the origin.

7. Since $y'_+(0) = \lim_{h\to 0^+} \frac{\sqrt[3]{h^2} - 0}{h} = \lim_{h\to 0^+} \sqrt[3]{\frac{h^2}{h^3}} = \lim_{h\to 0^+} \sqrt[3]{\frac{1}{h}} = \infty$,

the function $y=\sqrt[3]{x^2}$ is not differentiable at $x=0$. The graph of this function is as below:



8. (a)



(b) Since $-x^2 \le x^2 \sin(\frac{\pi}{x}) \le x^2$, $\lim_{x\to 0} g(x)-g(0) = \lim_{x\to 0} x^2 \sin(\frac{\pi}{x}) = 0$ by squeez theorem. Hence $g(x)$ is continuous at $x = 0$.

(c) Since $-|h| \le h\sin(\frac{\pi}{h}) \le |h|$, $g'(0) = \lim_{h\to 0} \frac{g(0+h)-g(0)}{h} = \lim_{h\to 0} \frac{h^2 \sin(\frac{\pi}{h})}{h}$

$= \lim_{h\to 0} h\sin(\frac{\pi}{h}) = 0$ by squeez theorem. Hence g is differentiable at $x = 0$ and $g'(0) = 0$.

9. Proof: Since $f(x)$ is even, $f(x) = f(-x)$ for all x in the domain. Consider $f_+'(0)$ and $f_-'(0)$.

$$f_+'(0) = \lim_{h\to 0^+} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0^+} \frac{f(h)-f(0)}{h} = -\lim_{h\to 0^+} \frac{f(-h)-f(0)}{-h}$$

$$\overset{(h'=-h)}{=} -\lim_{h'\to 0^-} \frac{f(h')-f(0)}{h'} = -f_-'(0).$$

Also $f(x)$ is differentiable at $x=0$, which indicates that $f_+'(0) = f_-'(0)$. So $f'(0) = f_+'(0) = f_-'(0) = 0$. $f'(0) = 0$ implies that the line tagent to the graph of f at $x=0$ is horizontal. □

Bonus Questions:

1. Proof: $f_+'(a) = \lim_{h\to 0^+} \frac{|a+h-a|\varphi(a+h)-0}{h} = \lim_{h\to 0^+} \frac{h\,\varphi(a+h)}{h} = \varphi(a)$

$f_-'(a) = \lim_{h\to 0^-} \frac{|a+h-a|\varphi(a+h)-0}{h} = \lim_{h\to 0^-} -\frac{h\,\varphi(a+h)}{h} = -\varphi(a)$

Since $\varphi(a) \neq 0$, $f'_+(a) \neq f'_-(a)$, which shows that f is not differentiable at $x = a$. □

2. Proof: We first show $\lim_{h \to 0} \frac{f(0+h) - f(h)}{h} = 0$. Note that for $h \neq 0$, $\left| \frac{f(0+h) - f(0)}{h} - 0 \right| = \left| \frac{f(h)}{h} \right| \leq \left| \frac{h^2}{h} \right| = |h|$. We choose $\delta := \varepsilon$. Then for $0 < |h - 0| < \delta$, $\left| \frac{f(0+h) - f(0)}{h} - 0 \right| \leq |h| < \varepsilon$. So $f'(x) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h} = 0$. $f(x)$ is differentiable at $x = 0$. Now consider other points excluding $x = 0$. For arbitrary $x \neq 0$, we choose $x_n \xrightarrow{n \to \infty} x$ and $y_n \xrightarrow{n \to \infty} x$, where x_n is rational and y_n is irrational for all n. Then $\lim_{n \to \infty} f(x_n) = \lim_{n \to \infty} x_n^2 = x^2$ and $\lim_{n \to \infty} f(y_n) = \lim_{n \to \infty} 0 = 0 \neq x^2$. So $\lim_{y \to x} f(y)$ do not exist and $f(x)$ is not continuous at x. Because x is an arbitrary point in $\mathbb{R} \setminus \{0\}$, $f(x)$ is not continuous at any points except $x = 0$. According to the statement in Question 5, $f(x)$ is not differentiable at these points. In conclusion, $f(x)$ is only differentiable at $x = 0$.

□